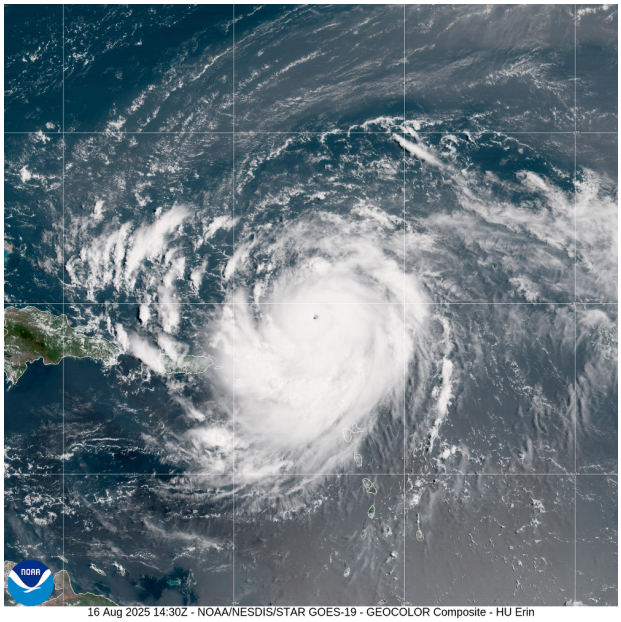
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

HURRICANE ERIN

(AL052025)

11–22 August 2025

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National Hurricane Center
30 January 2026[1]



GOES-19 GEOCOLOR IMAGE OF HURRICANE ERIN NEARING PEAK INTENSITY AT 1430 UTC 16 AUGUST 2025. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Erin was a very large and long-lived Cabo Verde hurricane that reached category 5 intensity (on the Saffir-Simpson Hurricane Wind Scale). It moved on a long parabolic track around the North Atlantic Ocean and produced wave and rip current impacts along a large stretch of the Atlantic coastline, causing 13 direct deaths.

[1] Original report dated 16 January 2026. This version corrects the OFCL (NHC) intensity forecast verification in the Forecast and Warning Critique section and verification tables (6a and 6b) due to an error that was found in the ATCF "a-deck" for Erin.

Hurricane Erin

11–22 AUGUST 2025

SYNOPTIC HISTORY

Erin originated from a tropical wave that moved off the west coast of Africa on 8 August. Over the next couple of days, an area of low pressure associated with the wave moved slowly westward to northwestward from the African coast to the Cabo Verde Islands. When the system neared the eastern Cabo Verde Islands, a consolidated area of deep convection formed, and the system acquired a sufficiently well-defined circulation to be considered as a tropical depression while located just southwest of the island of Sal around 0000 UTC 11 August. The system moved slightly north of west and became a tropical storm while making landfall on the island of Santo Antão in the northwestern Cabo Verde Islands by 0600 UTC 11 August. The "best track" chart of Erin's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[2].

After initial slight strengthening to an intensity of near 40 kt, Erin moved westward to west-southwestward across the tropical Atlantic from 11 to 13 August with little change in strength, likely due to relatively dry air and marginal sea surface temperatures (SSTs). A well-defined mid-level ridge to the north of the tropical cyclone maintained a mainly westward course during the early part of Erin's lifetime. The storm had strengthened only a little until 14 August, when the system's deep convection increased in coverage and organization. By 1200 UTC 15 August, Erin intensified into a hurricane while centered about 375 n mi east of the northernmost Leeward Islands. A slight weakness in the ridge resulted in a turn toward the west-northwest. Late on the 15th and early on the 16th, the hurricane turned back toward the west and, while moving over warmer ocean waters, began to intensify rapidly. Over the 12-hour period from 0600 to 1800 UTC 16 August, Erin's maximum winds increased by 45 kt and its central pressure deepened by 4.5 mb per hour. The hurricane reached a peak intensity of 140 kt, category 5 intensity on the Saffir-Simpson Hurricane Wind Scale, with a minimum central pressure of about 913 mb around 1800 UTC that day while its well-defined eye was centered about 80 n mi northeast of Anegada in the northernmost British Virgin Islands. Around that same time, aircraft observations indicated that Erin developed concentric eyewalls, and the hurricane began to weaken later on the 16th. The outer portions of Erin's southern circulation swept over the British and U.S. Virgin Islands and Puerto Rico, bringing tropical-storm-force winds to some of those areas. On 17 August, while passing a little over 100 n mi north of Puerto Rico, the system turned toward the west-northwest as it headed for a weakness in the subtropical ridge. The eye became less defined on satellite images, and Erin's maximum winds decreased to near 110 kt. The hurricane turned northwestward on 18 August, and its eye became better defined. Reports from Hurricane Hunter

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

aircraft indicated that the system had re-strengthened to a second peak intensity of near 120 kt around 1200 UTC 18 August, while centered about 90 n mi north-northeast of Grand Turk. The weakening and re-strengthening of the hurricane from 16 to 18 August coincided quite well with the dissipation of the inner eyewall and subsequent eyewall replacement during that period, as illustrated in Fig. 4. Erin caused some gusty winds over the Turks and Caicos Islands and portions of the southeastern Bahamas on 18–19 August.

After reaching its second peak in intensity, Erin turned north-northwestward, moving along the western periphery of a mid-tropospheric subtropical anticyclone. The hurricane weakened on 19 August due to increased westerly vertical wind shear, which resulted in the system's cloud pattern becoming more asymmetric. Erin's maximum winds decreased to near 85 kt by 1200 UTC that day. However, the shear soon diminished, the cloud pattern became more symmetric, and the hurricane strengthened again. Erin reached a third intensity peak of 95 kt by 1200 UTC 20 August while it was centered about midway between the Florida peninsula and Bermuda, while its circulation continued to expand. Hurricane Hunter observations indicated that the strongest winds were now located well east of the center. Erin turned northward while continuing to move on the western side of the anticyclone. The system then turned north-northeastward and northeastward with a gradual increase in forward speed on 21 August, passing about 175 n mi southeast of Cape Hatteras around 0600 UTC that day. The outer portion of Erin's western circulation brought tropical storm conditions to parts of the North Carolina and mid-Atlantic coastline. The cyclone weakened very slowly and accelerated northeastward to east-northeastward as it neared the mid-latitude westerlies. On 22 August, the system's cloud pattern took on a frontal appearance, and Erin became a hurricane-force extratropical cyclone around 1800 UTC that day. From 23 to 24 August, the system moved even faster toward the northeast over the northern Atlantic while maintaining its hurricane-force intensity. Later on the 24th and on 25-26 August, the cyclone moved much more slowly while making a counterclockwise loop a few hundred miles south of Iceland while weakening. The system merged with another extratropical low between Iceland and Ireland on 27 August.

METEOROLOGICAL STATISTICS

Observations in Erin (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Observations also include flight-level and dropwindsonde observations from flights of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command (AFRES) and NOAA. Figure 5 displays the aircraft missions into Erin. There were 24 aircraft missions tasked for Erin, 15 for AFRES and 9 for NOAA including 4 synoptic surveillance missions. Additionally, there were 4 research missions conducted by NOAA aircraft. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and

storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Erin.

Erin's wind field expanded considerably during its trek over the Atlantic, particularly after the cyclone moved into the subtropics. Erin's size ranged from the 92nd to the 95th percentile of all hurricanes on record in the Atlantic basin north of 20°N latitude.

Figure 6 shows the evolution of the tropical-storm-force wind field during the lifetime of Erin. Remarkably, in spite of its huge size, only the outer fringes of Erin's circulation passed over a few land locations after the system departed the Cabo Verde Islands.

Ship reports of winds of tropical storm force associated with Erin are given in Table 2, and selected surface observations from land stations and data buoys are given in Table 3. The highest wind report from a land station was a sustained 53-kt wind with a gust to 64 kt from Anegada, the northernmost of the British Virgin Islands.

Winds and Pressure

Erin's peak intensity of 140 kt at 1800 UTC 16 August is based on a maximum 700 mb flight-level wind of 154 kt from the AFRES at 1639 UTC 16 August. There was a surface (10 m) wind of 179 kt measured by a NOAA dropsonde at around 1403 UTC on the 16th, but this was apparently a gust since the low-level averaged wind from that drop corresponded to a sustained surface wind of around 129 kt. Erin's minimum central pressure of 913 mb is based on an AFRES dropsonde measurement in the eye of 919 mb around the same time with a surface wind of 59 kt.

Tropical-storm-force winds, either sustained or in gusts, were observed over portions of the British and U.S. Virgin Islands, in some locations along the North Carolina, Virginia, Maryland, Delaware, and New Jersey coasts, and on Bermuda.

Rainfall and Flooding

Two locations on São Vicente in the Cabo Verde Islands, Calhau and Mindelo, measured rainfall totals of 287 mm (11.30 in) and 224 mm (8.82 in) respectively from 0100 to 0800 UTC on 11 August. These rains caused significant freshwater flooding and the drowning deaths of 7 people on São Vicente. Rainfall totals of over 10 inches were observed in the British Virgin Islands (Table 3) with lesser amounts across the other islands of the northeastern Caribbean. Widespread flooding was reported across the British Virgin Islands. Significant impacts included extensive landslides and considerable damage to the road network on Virgin Gorda. While no lives were lost, ten residents sought shelter in Anegada.

Storm Surge

Erin produced minor storm surge flooding accompanied by damaging waves along the Outer Banks of North Carolina, the shores of the lower Chesapeake Bay, and the Delmarva Peninsula, as well as the shores of the Delaware Bay. Maximum storm surge inundations of 1 to 3 feet above ground level (AGL) were widespread along the coast from North Carolina to New Jersey due to Erin's notably large wave field. Figure 7 shows the National Ocean Service (NOS)

tide station observations along the Mid-Atlantic coast. The NOS tide gauge in Duck, North Carolina, measured a peak water level of 3.4 ft Mean Higher High Water (MHHW). Elsewhere, observations peaked at 3.3 ft MHHW in Wachapreague, Virginia, 3.0 ft MHHW in Lewes, Delaware, and 3.0 ft MHHW in Cape May, New Jersey.

CASUALTY AND DAMAGE STATISTICS

Thirteen fatalities are directly attributed to Erin[3]. There were seven deaths due to drownings from freshwater flooding reported on São Vicente in the Cabo Verde Islands. Three persons drowned due to rip currents in Puerto Rico. A person drowned, presumably due to rip currents, at La Romana in the eastern Dominican Republic. There was also one rip current drowning at Sailors Haven on Fire Island, New York, and one rip current drowning at Hampton Beach, New Hampshire.

On the North Carolina Outer Banks, extremely large surf on top of storm surge breached dune structures in certain vulnerable areas, covering roads with sand and water. Two structures in Buxton sustained major damage. There was flooding that caused the closure of North Carolina Highway 12 (which runs along the Outer Banks) at Oregon Inlet. The total damage estimate was around $3.1 million from North Carolina, mostly in the community of Buxton in Dare County.

Dozens of people had to be rescued from stranded vehicles and restaurants as floodwaters rose in Margate City on the Jersey Shore.

Impacts to Bermuda were minor, with minimal power outages across the island.

FORECAST AND WARNING CRITIQUE

Erin's genesis was not anticipated very far in advance (Table 4), partly because the system developed very far east in the Atlantic basin. The wave from which Erin developed was introduced into the Tropical Weather Outlook (TWO) when it was near the west coast of Africa 48 h prior to genesis with a low (<40%) chance for development within 7 days. The 7-day formation probability was raised to medium (40%-60%) 36 hours prior to the time of genesis and raised to high (>60%) just 18 hours before genesis. The 2-day formation probability was set to low 30 hours prior to genesis and raised to medium only 10.5 hours before genesis, in a special TWO. The 2-day formation probability was never raised to high before the system formed. The location of Erin's genesis was also not well forecast in the Graphical Tropical Weather Outlooks with the anticipated

[3] Deaths occurring as a direct result of the forces of the tropical cyclone are referred to as "direct" deaths. These would include those persons who drowned in storm surge, rough seas, rip currents, and freshwater floods. Direct deaths also include casualties resulting from lightning and wind-related events (e.g., collapsing structures). Deaths occurring from such factors as heart attacks, house fires, electrocutions from downed power lines, vehicle accidents on wet roads, etc., are considered "indirect" deaths.

development area over the tropical Atlantic too far to the west of the actual formation point (Fig. 8).

A verification of NHC official track forecasts for Erin, which were mostly quite good, is given in Table 5a. Official track forecast errors were lower than or comparable to the mean official errors for the previous 5-yr period for all forecast intervals. A homogeneous comparison of the official track errors with selected guidance models is given in Table 5b. The HCCA and Google DeepMind (GDMI) guidance were the best performers. Most of the official forecasts were along or slightly east of Erin's observed track (not shown).

A verification of NHC official intensity forecasts for Erin is given in Table 6a. At all forecast periods, the mean official intensity forecast errors were higher than the mean official errors for the previous 5-yr period. The mean climatology and persistence intensity forecast (OCD5) errors for Erin were comparable to the 5-yr averages. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 6b. For the 24-h through 120-h forecast intervals, the GDMI had the lowest mean errors. The rapid intensification of Erin on 16 August was not very well predicted in the official forecasts displayed in Fig. 9.

Watches and warnings associated with Erin are listed in Table 7. Although the hurricane was not expected to make landfall, the proximity of Erin's large outer wind field prompted the issuance of Tropical Storm Watches and/or Warnings for portions of the Leeward Islands, the Southeast and Central Bahamas, the North Carolina and Virginia coasts, and Bermuda.

The NWS issued storm surge watches and warnings for very exposed portions of the North Carolina coast due to the potential for coastal inundation from Erin. A Storm Surge Watch was first issued at 2100 UTC 18 August from Cape Lookout to Duck, North Carolina. The peak storm surge forecast was 2 to 4 ft AGL. The area within the Storm Surge Watch was upgraded to a Storm Surge Warning at 1500 UTC 19 August. Due to Erin's proximity to the coast, destructive wave action was forecast to cause significant beach erosion, leading to coastal flooding, rendering roads in the Outer Banks of North Carolina impassable. This prompted evacuation decisions in Dare County, North Carolina. The NOS observation at Duck, NC, fell within the forecast range.

Outside of the Storm Surge Warning area, the peak storm surge forecast was 1 to 3 feet from Cape Fear, North Carolina, to Montauk Point, New York. Peak water level observations from NOS stations were largely up to 3 ft along the Mid-Atlantic and Northeast coastlines (Fig. 7).

IMPACT-BASED DECISION SUPPORT SERVICES (IDSS) AND PUBLIC COMMUNICATION

NHC began formal communication with emergency managers on Friday, August 15 as Erin was strengthening east of the Caribbean. Five decision support briefings were provided to emergency managers and coordinated through the FEMA Hurricane Liaison Team embedded at the NHC. The briefings were federal video-teleconferences with FEMA HQ and FEMA Region 2, 3, and 4. These briefings continued through Friday, August 21, until Erin started moving away from the United States.

ACKNOWLEDGEMENTS

Ester Araújo de Brito provided rainfall data from Cabo Verde. We thank Ian Currie of the Bermuda Weather Service for providing detailed observations recorded on the island. Lisa Bucci of the NHC produced Figs. 5 and 9, Cassandra Mora of the NHC provided storm surge information and produced Figure 7, and Philippe Papin of the NHC produced Figs. 4 and 8.

Table 1. Best track for Hurricane Erin, 11–22 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
11 / 0000	16.5	23.1	1007	30	tropical depression
11 / 0600	17.2	25.2	1006	35	tropical storm
11 / 1200	17.3	27.3	1004	40	"
11 / 1800	17.4	29.3	1004	40	"
12 / 0000	17.5	31.3	1004	40	"
12 / 0600	17.5	33.3	1004	40	"
12 / 1200	17.3	35.4	1004	40	"
12 / 1800	17.1	37.4	1004	40	"
13 / 0000	16.8	39.3	1004	40	"
13 / 0600	16.6	41.0	1004	40	"
13 / 1200	16.4	42.7	1002	45	"
13 / 1800	16.3	44.3	1002	45	"
14 / 0000	16.3	45.9	1002	45	"
14 / 0600	16.3	47.5	1002	45	"
14 / 1200	16.3	48.9	999	50	"
14 / 1800	16.6	50.4	998	55	"
15 / 0000	16.9	51.9	997	60	"
15 / 0600	17.4	53.5	997	60	"
15 / 1200	18.0	55.2	996	65	hurricane
15 / 1800	18.7	56.8	993	65	"
16 / 0000	19.4	58.5	981	75	"
16 / 0600	19.6	60.4	967	95	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
16 / 1200	19.6	62.1	934	130	"
16 / 1800	19.8	63.4	913	140	"
17 / 0000	20.0	64.6	934	125	"
17 / 0600	20.3	65.8	940	110	"
17 / 1200	20.7	67.0	940	110	"
17 / 1800	21.5	68.0	946	110	"
18 / 0000	22.0	68.9	946	110	"
18 / 0600	22.5	69.7	945	115	"
18 / 1200	22.9	70.5	935	120	"
18 / 1800	23.5	71.0	937	115	"
19 / 0000	24.0	71.4	947	100	"
19 / 0600	24.5	71.8	955	90	"
19 / 1200	25.2	72.1	958	85	"
19 / 1800	26.1	72.5	958	85	"
20 / 0000	27.2	72.8	959	85	"
20 / 0600	28.3	73.2	954	90	"
20 / 1200	29.5	73.7	943	95	"
20 / 1800	30.7	73.6	942	95	"
21 / 0000	32.1	73.3	941	95	"
21 / 0600	33.6	72.6	945	90	"
21 / 1200	34.9	71.7	949	90	"
21 / 1800	35.9	70.1	952	85	"
22 / 0000	36.9	68.4	954	85	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
22 / 0600	37.9	66.5	956	80	"
22 / 1200	38.5	64.6	956	80	"
22 / 1800	39.4	61.4	957	80	extratropical
23 / 0000	40.4	58.1	957	80	"
23 / 0600	41.3	54.7	962	75	"
23 / 1200	42.0	50.5	962	75	"
23 / 1800	43.0	45.0	962	70	"
24 / 0000	45.5	40.5	961	70	"
24 / 0600	48.0	35.3	960	70	"
24 / 1200	50.8	30.9	952	80	"
24 / 1800	52.6	28.2	952	80	"
25 / 0000	53.7	26.4	954	70	"
25 / 0600	54.9	23.9	955	60	"
25 / 1200	56.1	22.1	957	55	"
25 / 1800	57.3	21.3	958	50	"
26 / 0000	58.3	20.9	961	45	"
26 / 0600	59.3	20.6	962	45	"
26 / 1200	59.9	21.6	963	45	"
26 / 1800	60.3	22.5	963	45	"
27 / 0000	60.0	23.3	968	45	"
27 / 0600	59.3	22.7	975	40	"
27 / 1200	58.6	20.9	980	40	"
27 / 1800	58.0	19.0	985	40	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
28 / 0000					merged with extratropical low
16 / 1800	19.8	63.4	913	140	Maximum winds and minimum pressure
11 / 0600	17.2	25.2	1006	35	Landfall in Santo Antão, Cabo Verde

Table 2. Selected ship reports with winds of at least 34 kt for Hurricane Erin, 11–22 August 2025.

Date/Time (UTC)	Ship call sign	Latitude (°N)	Longitude (°W)	Wind dir/ speed (kt)	Pressure (mb)
21/1600	ATR9M5	40.6	72.2	020 / 40	1013.8

Table 3. Selected surface observations for Hurricane Erin, 11-22 August 2025.

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
UNITED STATES									
International Civil Aviation Organization (ICAO) Sites									
North Carolina									
Manteo Airport (KMQI) (35.90N, 75.70W)	21/0825	1002.4	21/1005	28	38				
Billy Mitchell Field (KHSE) (35.22N, 75.62W)	21/0751	999.5	21/0835	19	37				
Beaufort Airport (KMRH) (34.72N, 76.65W)	21/0758	1002.5	21/0910	22	34				
Dare County Gunnery Range (K2DP) (35.67N, 75.90W)	21/0755	1002.0	21/1055	23	37				
Piney Island - BT11 Bombing Range (KNBT) (35.02N, 76.46W)	21/0751	1002.5	21/1128	25	37				
Elizabeth City CGAS (KECG) (36.26N, 76.17W)	21/0854	1004.2	21/1454	24	35				
Virginia									
Oceana NAS (KNTU) (36.82N, 76.03W)	21/0856	1005.7	21/1456	23	38				
Norfolk Intl (KORF) (36.90N, 76.19W)	21/0851	1006.2	21/1302	24	38				
Wallops Island NASA Test Facility (KWAL) (37.94N, 75.47W)	21/0854	1007.7	21/1605	25	38				
Maryland									
Ocean City (KOXB) (38.31N, 75.12W)	21/1053	1008.6	21/1510	23	39				
Puerto Rico									
San Juan (TJSJ) (18.44N, 66.00W)	17/0656	1003.7	16/2350	27	39				
U.S. Virgin Islands									
St. Thomas (TIST) (18.34N, 64.97W)	16/2153	1003.2	17/0550	28	50				
Public/Other									
North Carolina									
Buxton (XBUX; WeatherFlow) (35.26N, 75.52W)	21/0801	997.2	21/1251	23 (10.4 m)	38				
Avon-Sound (XAVN; WeatherFlow) (35.37N, 75.51W)	21/0826	999.8	21/0851	34 (4.6 m)	43				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Waves (XWAV; WeatherFlow) (35.57N, 75.47W)	21/0844	998.5	21/1404	30 (10.7 m)	41				
Oregon Inlet (XORI; WeatherFlow) (35.80N, 75.55W)	21/0811	998.9	21/0851	29 (10.4 m)	40				
Hatteras High (XHAT; WeatherFlow) (35.26N, 75.55W)	21/0756	997.9	21/0516	30 (10.4 m)	42				
Croatan Sound (XCTNr; WeatherFlow) (35.87N, 75.66W)	21/0820	1001	21/1230	25 (7.9 m)	40				
Real Slick (XSLK; WeatherFlow) (35.57N, 75.49W)	21/1452	1003.5	21/1452	32 (6.1 m)	40				
Pamlico Sound (XPM2; WeatherFlow) (35.43N, 75.83W)	21/0820	999	21/0940	35 (13.1 m)	44				
Ocracoke Sound (XOCR; WeatherFlow) (35.13N, 76.01W)	21/0828	1000	21/1048	32 (7.6 m)	40				
Avon-Ocean (XAVO; WeatherFlow) (35.35N, 75.50W)	21/0813	999	21/0918	31 (11.9 m)	42				
Alligator River Bridge (XALI; WeatherFlow) (35.90N, 76.01W)	21/0844	1003	21/0924	36 (11.6 m)	44				
Kites Resort (XRTH; WeatherFlow) (35.58N, 75.47W)	21/0804	998	21/1719	29 (16.1 m)	40				
Jennette's Pier (XJNP; WeatherFlow) (35.91N, 75.59W)	21/0815	1002	21/1215	40 (18.3 m)	48				
Virginia									
Sandbridge (XSRB; WeatherFlow) (36.70N, 75.93W)	21/0850	1005	21/1350	24 (4.6 m)	34				
New Point Comfort (XNPC; WeatherFlow) (37.33N, 76.27W)	21/0941	1005.3	21/1231	28 (12.8 m)	38				
Thimble Shoals (XTHM; WeatherFlow) (37.05N, 76.26W)	21/0940	1005.5	21/1315	28 (6.1 m)	36				
Rudee Inlet (XRUD; WeatherFlow) (36.83N, 75.97W)			21/1112	29 (8.8 m)	35				
Lynnhaven Inlet Light 2 (XLHI; WeatherFlow) (36.92N, 76.09W)	21/0924	1005	21/1439	28 (6.6 m)	34				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Silver Beach (XSIL; WeatherFlow) (37.49N, 75.97W)	21/0911	1005	21/1626	29 (7.6 m)	35				
Middle Ground Lighthouse (XMGL; WeatherFlow) (36.945N, 76.392W)	21/0811	1002.9	21/1126	28 (19.7 m)	37				
Cape Henry (XHEN; WeatherFlow) (36.927N, 76.007W)			21/1028	33 (22.9 m)	41				
Maryland									
Ocean City (XOCN; WeatherFlow) (38.33N 75.08W)	21/1648	1006	21/1738	37 (10.4 m)	43				
Ocean City Drawbridge (MD151; Maryland DOT) (38.33N 75.09W)			21/1234	31	46				
Plantation Flats (XPLT; WeatherFlow) (37.243N, 76.029W)			21/1545	31 (13.1 m)	36				
Delaware									
Indian River Inlet (DEL18; Delaware DOT) (38.61N 75.06W)			21/1642	34	41				
New Jersey									
Harvey Cedars (NJ11; NJ WeatherNet) (39.70N 74.14W)			21/1105	35	37				
U.S. Virgin Islands									
Buck Island (XBUK) (18.28N, 64.89W)			17/0552	40	50				
Two Brothers (XBRO) (18.34N, 64.82W)			17/0603	35	45				
St. Thomas (18.34N 64.97W)			17/0545	28	44				
Puerto Rico									
San Juan (XJUA) (18.48N, 66.10W)	17/0602	1001.5	16/2347	29	36				
Luquillo			17/0025	23	39				
Isla Culebrita Light (XCUL) (18.31N, 65.23W)			17/0645	40	49				
BRITISH VIRGIN ISLANDS									
Surface Stations									
The Settlement, Anegada (Datascope/WeatherFlow)			16/2330	53	64				
Anegada	16/2200	1001.7	17/0119	44					6.78
Georges North Side			16/1926	36					9.51

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Brewers Bay	16/2200	1003.5	17/0101	38					7.98
Cane Garden Bay	M	1005.8	17/0531	25					9.14
Balsum Ghut			16/2300	47					7.67
Hodges Creek	16/2100	994	17/0618	43					10.19
Paraquita Bay			17/0036	43					9.91
East End	17/0000	1003.2	17/0600	35					9.71
Kingston	16/2200	1006.3	16/2124	31					9.72
West End	M	1004.4	16/2239	34					6.20
Road Town			16/2054	20					9.79
Chalwell			17/0544	32					7.78
Hannahs	16/2200	1001.2	17/0122	32					8.61
Jost Van Dyke	17/1200	1004.1	17/1355	31					7.00
Tortola (1456W)			17/0550	25	50				
SAINT-MARTIN AND SAINT-BARTHÉLEMY									
International Civil Aviation Organization (ICAO) Sites									
Saint-Martin Grand Case Airport (TFFG) (18.00N, 63.05W)	16/1900	1006.3	16/2300	22	55				2.24
Saint-Barthélemy Airport (TFFJ) (17.90N, 62.84W)			16/1900	23	41				
Surface Stations									
Saint-Barthélemy Gustavia Station (17.90N, 62.85W)	16/2000	1005.4	16/2100	40	54				1.29
TURKS AND CAICOS ISLANDS AND THE BAHAMAS									
International Civil Aviation Organization (ICAO) Sites									
Providenciales (MBPV) (21.77N, 72.27W)			19/0150	23	46				
San Salvador (MYSM) (24.06N, 74.52W)			18/2131	19	30				
Cockburn Town, Grand Turk (MBGT) (21.44N, 71.14W) *Data Incomplete			Missing	18	30				
BERMUDA									
International Civil Aviation Organization (ICAO) Sites									
L. F. Wade International Airport (TXKF) (32.36N, 64.68W)	22/0720	1008.1	22/2145	34 (10-min)	50				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Surface Stations									
National Museum of Bermuda (NMB) (32.33N, 64.83W)	22/0650	1005.2	21/2146	38 (46 m)	51				
AviMet 30 End of TXKF Runway 30 (32.36N, 64.67W)	22/0714	1007.9	21/2224	43	50				
AviMet 12 End of TXKF Runway 12 (32.37N, 64.69W)	22/0652	1007.6	21/2223	36	43				
ESSO Pier (32.37N, 64.70W)	22/0712	1008.1	22/0812	32 (4.6 m)	44				
MARINE OBSERVATIONS									
National Ocean Service (NOS) Sites									
Coastal New Jersey									
Cape May, NJ (CMAN4) (38.97N, 74.96W)						2.85	5.44	3.01	
Ship John Shoal, NJ (SJSN4) (39.31N, 75.38W)								2.81	
Coastal Delaware									
Brandywine Shoal Light, DE (BRND1) (38.99N, 75.11W)	21/1054	1010.5	21/1436	33 (21.1 m)	39			2.83	
Lewes, DE (LWSD1) (38.78N, 75.12W)	21/1018	1010	21/1606	30 (9.5 m)	37	2.96	5.08	3.05	
Coastal Maryland									
Ocean City Inlet, MD (OCIM2) (38.33N, 75.09W)	21/1036	1008.3	21/1912	22 (8.5 m)	36	2.58	3.13	2.41	
Coastal Virginia									
Chesapeake Bay Bridge Tunnel, VA (CHBV2) (37.03N, 76.08W)	21/0912	1005.2	21/1812	29 (4.1 m)	34			2.95	
Kiptopeke, VA (KPTV2) (37.17N, 75.99W)	21/0906	1007.3	21/1412	21 (7.2 m)	27	2.37	3.57	2.53	
Yorktown USCG Training Center, VA (YKTV2) (37.23N, 76.48W)						2.35	3.44	2.39	
Wachapreague, VA (WAHV2) (37.61N, 75.69W)	21/0954	1007.3	21/0806	22 (7.0 m)	29	3.36	5.15	3.29	
Willoughby Degaussing Station, VA (WDSV2) (36.98N, 76.32W)	21/0948	1006.6	21/1236	28 (10.0 m)	34				
Sewells Point, VA (SWPV2) (36.94N, 76.33W)	21/0954	1006.2				2.67	3.92	2.78	

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Cape Henry, VA (CHYV2) (36.93N, 76.01W)	21/0912	1004.1	21/1024	34 (22.9 m)	38				
Money Point, VA (MNPV2) (36.78N, 76.30W)	21/0924	1005.8	21/1318	17 (5.8 m)	25			2.74	
Rappahannock Light, VA (RPLV2) (37.54N, 76.02W)	21/0954	1008	21/1700	32 (5.0 m)	37				
Coastal North Carolina									
Duck, NC (DUKN7) (36.18N, 75.75W)	21/0806	1002.6	21/0900	34 (8.6 m)	40	3.08	4.85	3.36	
Oregon Inlet, NC (ORIN7) (35.77N, 75.53W)	21/0812	1000.6	21/1118	32 (6.7 m)	42				
USCG Hatteras, NC (HCGN7) (35.21N, 75.70W)	21/0830	1000	21/1036	36 (7.6 m)	46	2.45	2.65	2.39	
Beaufort Duke Marine Lab, NC (BFTN7) (34.72N, 76.67W)						1.89	3.51	2.05	
Wilmington, NC (WLON7) (34.23N, 77.95W)						1.9	4.13	2.04	
Wrightsville Beach, NC (JMPN7) (34.21N, 77.79W)						2.06	4.18	2.45	
Coastal U.S. Virgin Islands									
Charlotte Amalie, USVI (CHAV3) (18.34N, 64.92W)	16/2200	1003.1	17/0548	33 (6.5 m)	44				
Coastal Puerto Rico									
San Juan, La Puntilla, San Juan Bay, PR (SJNP4) (18.46N, 66.12W)	17/0600	1003.5	16/2348	31 (5.7 m)	42				
Buoys									
44058 Stingray Point, VA (IOOS Partners) (37.57N, 76.26N)			21/1812	25	35				
44072 York Spit, VA (IOOS Partners) (37.20N, 76.27N)			21/1548	27	35				
41064 Onslow Bay, NC (CORMP) (34.21N, 76.95N)	21/0808	1003.7	21/0508	25	37				
41025 Diamond Shoals (NOAA/NDBC) (35.01N, 75.40W)	21/0830	996.9	21/0915	42 (1-min)	49				
41052 S of St. John, USVI (CarICOOS) (18.25N, 64.76N)	17/0610	1003	17/0540	29 (1-min)	43				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
41056 Vieques Island, PR (18.26N, 65.46N)	17/0710	1003.1	17/0820	27 (1-min)	39				
41053 San Juan, PR (CarICOOS) (18.47N, 66.10N)	17/0600	1002	16/2340	25 (1-min)	41				
41043 NE Puerto Rico (NOAA/NDBC) (21.09N, 64.86N)	17/0300	996.2	17/0222	54 (1-min)	65				
41046 E Bahamas (NOAA/NDBC) (23.84N, 68.34N)	18/0600	1002	18/0623	48 (1-min)	55				
41047 NE Bahamas (NOAA/NDBC) (27.56N, 71.48N) *Winds failed 19/2220 UTC	19/2300	987	19/2203	58 (1-min)	68				
41002 S Hatteras (NOAA/NDBC) (31.74N, 74.96N)	20/2049	985.5	20/2216	51 (1-min)	67				
41001 E Hatteras (NOAA/NDBC) (34.79N, 72.42N)	21/1130	962.4	21/0725	60 (1-min)	73				
44014 Virginia Beach (NOAA/NDBC) (36.60N, 74.84N)	21/1100	1001	21/1727	34 (1-min)	43				
44008 Nantucket (NOAA/NDBC) (40.50N, 69.25N)	22/0611	1000.3	22/0245	38 (1-min)	47				
44020 Nantucket Sound (NOAA/NDBC) (41.50N, 70.28N)	22/0620	1007	22/0703	29 (1-min)	34				
44011 Georges Bank (NOAA/NDBC) (41.09N, 66.55N)	22/0940	996.4	22/0905	42 (1-min)	53				
44137 East Scotia Slope (Environment Canada) (42.26N, 62.03N)	22/1400	999.1	22/1400	37					
44139 Banqureau Banks (Environment Canada) (44.24N, 57.10N)	23/0500	1000	23/0500	37					
44150 La Have Bank (Environment Canada) (42.50N, 64.02N)	22/1600	1000.6	22/1600	35					
Coastal-Marine Automated Network (C-MAN) Sites									
CLKN7 Cape Lookout, NC (34.61N, 76.53N)	21/0717	1001.5	21/1405	27 (1-min)	33				
BUZM3 Buzzards Bay, MA (41.40N, 71.03N)	22/0700	1008.9	21/2020	33 (10-min)	38				
Public/Other									
Chesapeake Light Tower, VA (XCLT; WeatherFlow) (36.905N, 75.713W)	21/1035	1000	21/1710	39 (41.2 m)	45				

Location	Minimum Sea Level Pressure		Maximum Surface Wind Speed			Storm surge (ft)[c]	Storm tide (ft)[d]	Estimated Inundation (ft)[e]	Total rain (in)
	Date/ time (UTC)	Press. (mb)	Date/ time (UTC)[a]	Sustained (kt)[b]	Gust (kt)				
Tangier Sound, VA (XTSL; WeatherFlow) (37.788N, 75.974W)	21/0936	1006.7	21/1506	30 (16.5 m)	34				
Third Island CBBT (XBBT; WeatherFlow) (37.037N, 76.077W)			21/1808	35 (15.8 m)	43				

[a] Date/time is for sustained wind when both sustained and gust are listed.
[b] Except as noted, sustained wind averaging periods for C-MAN and land-based reports are 2 min; buoy averaging periods are 8 min.
[c] Storm surge is water height above normal astronomical tide level.
[d] For most locations, storm tide is water height above the North American Vertical Datum of 1988 (NAVD88). Storm tide is water height above Mean Lower Low Water (MLLW) for NOS stations in Puerto Rico, the U.S. Virgin Islands, and Barbados.
[e] Estimated inundation is the maximum height of water above ground. For some USGS storm tide pressure sensors, inundation is estimated by subtracting the elevation of the sensor from the recorded storm tide. For other USGS storm tide sensors and USGS high-water marks, inundation is estimated by subtracting the elevation of the land derived from a Digital Elevation Model (DEM) from the recorded and measured storm tide. For NOS tide gauges, the height of the water above Mean Higher High Water (MHHW) is used as a proxy for inundation.

Table 4. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	30	48
Medium (40%-60%)	10.5	36
High (>60%)	-	18

Table 5a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Erin, 11–22 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**16.2**	**29.5**	**42.7**	**57.4**	74.1	91.5	**115.2**	**173.8**
OCD5	33.2	76.2	118.8	155.1	189.7	222.2	300.1	401.5
Forecasts	43	41	39	37	35	33	29	25
OFCL (2020-24)	23.0	34.3	45.8	58.7	73.5	89.8	128.7	185.4
OCD5 (2020-24)	45.1	95.7	150.9	203.1	252.7	295.4	366.2	426.6

Table 5b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Erin, 11–22 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	16.5	30.1	43.1	56.6	73.0	90.0	115.1	174.2
OCD5	32.8	74.2	114.1	147.0	178.0	205.5	283.3	374.3
GFSI	21.3	42.4	59.6	78.6	103.9	130.7	181.2	277.6
HWFI	19.8	38.6	55.8	74.3	92.5	113.2	151.3	196.2
HMNI	21.1	41.2	61.9	78.3	98.2	114.2	142.0	192.4
HFAI	19.8	38.9	53.5	65.4	76.1	**83.9**	**97.5**	**131.9**
HFBI	22.0	42.5	59.4	73.1	88.8	97.2	**113.8**	**148.4**
GDMI	**15.5**	**24.0**	**33.0**	**44.1**	**59.0**	**74.2**	116.7	**161.5**
CTCI	24.2	48.1	69.4	88.2	112.7	138.9	177.0	216.1
EMXI	18.4	36.0	51.3	64.3	78.2	**83.0**	**97.2**	**124.2**
CMCI	21.3	39.9	59.3	76.0	92.6	108.3	149.3	198.5
NVGI	23.2	47.0	72.5	97.7	124.7	157.5	231.7	361.0
TVCA	18.6	36.1	52.1	64.8	80.6	95.5	121.4	**162.7**
TVCX	18.2	35.1	50.6	63.4	78.9	92.8	117.0	**154.8**
GFEX	19.0	36.5	52.1	67.2	86.0	103.2	132.7	188.2
TVDG	18.3	35.1	50.3	62.6	78.1	91.5	119.0	**161.1**
HCCA	**16.4**	**26.3**	**36.4**	**47.8**	**62.6**	**73.8**	**92.0**	**132.4**
FSSE	16.9	30.8	**42.0**	**54.8**	73.5	**88.1**	117.1	178.9
AEMI	20.7	40.2	58.3	73.8	93.0	110.5	138.2	192.0
TABS	37.8	80.5	112.4	137.1	151.2	150.4	145.4	215.4
TABM	26.3	58.0	83.5	106.2	130.5	151.8	172.2	254.5
TABD	22.3	48.3	81.6	121.3	166.2	208.9	282.6	387.3
Forecasts	41	39	37	35	33	31	28	24

Table 6a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Erin, 11–22 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	7.6	11.7	13.1	13.6	14.7	15.8	16.0	14.8
OCD5	9.3	13.2	14.6	16.8	19.4	20.7	20.6	19.6
Forecasts	43	41	39	37	35	33	29	25
OFCL (2020-24)	5.1	7.3	8.6	10.0	10.5	10.9	12.4	13.6
OCD5 (2020-24)	6.8	10.6	13.8	16.5	17.9	19.2	21.4	19.9

Table 6b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Erin, 11–22 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 6a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	7.6	11.9	13.2	13.7	14.5	15.5	15.7	14.2
OCD5	9.4	13.5	14.7	17.1	19.7	21.1	21.0	17.9
HWFI	10.0	13.9	16.2	15.1	15.2	**14.4**	**14.5**	16.0
HMNI	10.3	13.5	**12.9**	**12.7**	**13.8**	**14.5**	**14.2**	17.5
HFAI	9.9	14.4	18.0	19.1	17.2	15.9	**15.6**	15.9
HFBI	9.1	12.2	14.6	16.8	17.8	18.1	16.6	14.2
GDMI	8.2	**10.4**	**10.6**	**9.7**	**12.0**	**12.5**	**13.9**	**13.6**
CTCI	9.5	13.1	**12.4**	**13.4**	17.6	20.6	25.6	26.7
DSHP	8.7	12.5	14.2	16.8	20.0	23.5	24.0	21.8
LGEM	9.3	13.1	14.6	17.4	19.6	22.4	21.3	19.6
ICON	8.4	**11.4**	**12.5**	13.8	15.2	17.4	18.0	18.1
IVCN	8.0	**10.6**	**11.5**	**12.9**	**13.5**	**15.0**	17.1	17.1
IVDR	8.0	**10.8**	**11.5**	**12.8**	**13.2**	**14.5**	16.8	16.6
GFSI	9.9	13.7	16.6	18.2	20.3	22.0	23.7	17.1
EMXI	10.4	15.5	18.5	23.1	27.3	29.0	27.7	27.5
HCCA	8.2	**10.5**	**12.2**	14.1	14.5	**14.7**	16.0	15.1
FSSE	8.0	**10.3**	**12.1**	14.2	15.1	16.3	18.3	17.8
Forecasts	41	39	37	35	33	31	28	24

Table 7. Watch and warning summary for Hurricane Erin, 11–22 August 2025.

Date/Time (UTC)	Action	Location
14 / 2100	Tropical Storm Watch issued	Anguilla and Barbuda
14 / 2100	Tropical Storm Watch issued	St. Martin and St. Barthelemy
14 / 2100	Tropical Storm Watch issued	Saba and St. Eustatius
14 / 2100	Tropical Storm Watch issued	Sint Maarten
16 / 0000	Tropical Storm Watch discontinued	Barbuda
16 / 0600	Tropical Storm Watch discontinued	Saba and St. Eustatius
16 / 0900	Tropical Storm Watch discontinued	Anguilla
16 / 2100	Tropical Storm Watch issued	Turks and Caicos
17 / 0300	Tropical Storm Watch discontinued	St. Martin and St. Barthelemy
17 / 0300	Tropical Storm Watch discontinued	Sint Maarten
17 / 0900	Tropical Storm Watch changed to Tropical Storm Warning	Turks and Caicos
17 / 0900	Tropical Storm Watch issued	Southeast Bahamas
17 / 1800	Tropical Storm Watch changed to Tropical Storm Warning	Southeast Bahamas
18 / 0900	Tropical Storm Watch issued	Central Bahamas
18 / 2100	Storm Surge Watch issued	Cape Lookout to Duck
18 / 2100	Tropical Storm Watch issued	Beaufort Inlet to Duck
19 / 1500	Tropical Storm Watch changed to Tropical Storm Warning	Beaufort Inlet to Duck
19 / 1500	Storm Surge Watch changed to Storm Surge Warning	Cape Lookout to Duck
19 / 1500	Tropical Storm Watch issued	Duck to Cape Charles Light
19 / 1800	Tropical Storm Watch discontinued	Central Bahamas

Date/Time (UTC)	Action	Location
19 / 1800	Tropical Storm Warning discontinued	Turks and Caicos
19 / 1800	Tropical Storm Warning discontinued	Southeast Bahamas
19 / 2100	Tropical Storm Watch modified to	Duck to Chincoteague
19 / 2100	Tropical Storm Watch issued	Bermuda
20 / 0900	Tropical Storm Watch modified to	NC/VA Border to Chincoteague
20 / 0900	Tropical Storm Warning modified to	Beaufort Inlet to NC/VA Border
20 / 1500	Tropical Storm Watch discontinued	NC/VA Border to Chincoteague
20 / 1500	Tropical Storm Warning modified to	Beaufort Inlet to Chincoteague
21 / 1800	Tropical Storm Watch changed to Tropical Storm Warning	Bermuda
21 / 2100	Tropical Storm Warning modified to	Duck to Chincoteague
22 / 0000	Tropical Storm Warning discontinued	Duck to Chincoteague
22 / 0300	Storm Surge Warning discontinued	Cape Lookout to Duck
22 / 0900	Tropical Storm Warning discontinued	All

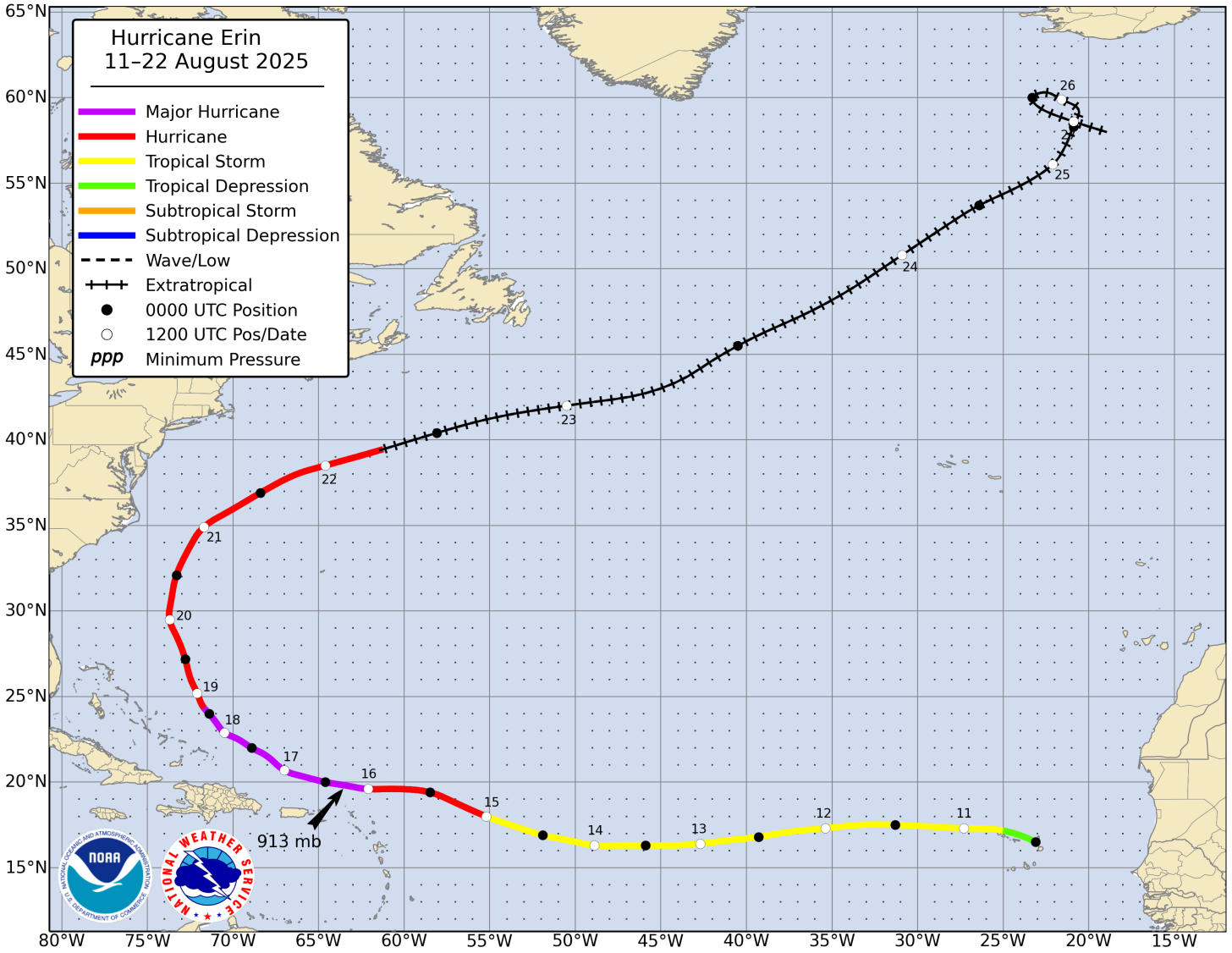


Figure 1. Best track positions for Hurricane Erin, 11–22 August 2025. Track during the extratropical stage is partially based on analyses from the NOAA Ocean Prediction Center.

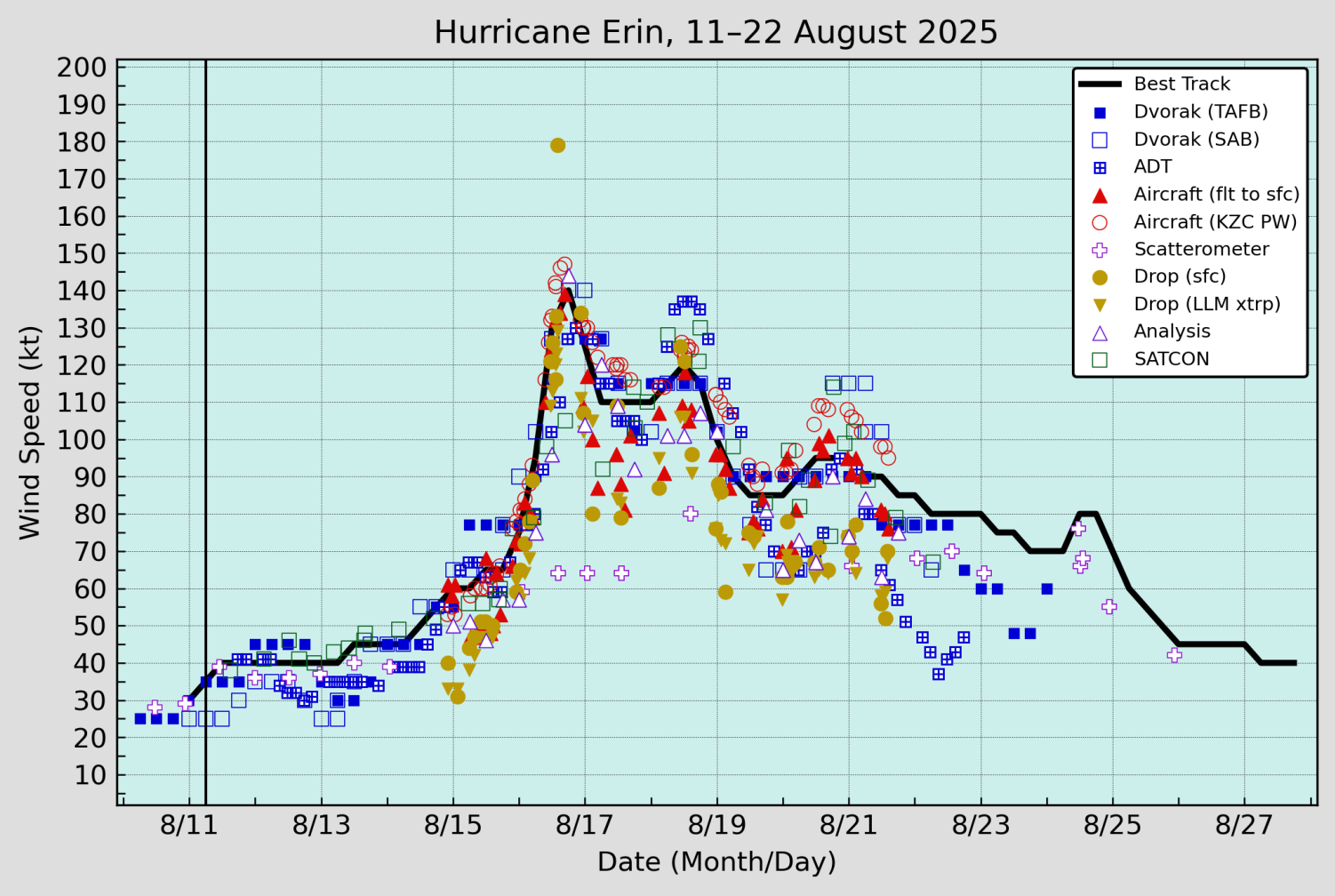


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Erin, 11–2 August 2025. Aircraft observations have been adjusted for elevation using 90%, 80%, and 75% adjustment factors for observations from 700 mb, 850 mb, and 925 mb, respectively. Dropwindsonde observations include actual 10 m winds (sfc), as well as surface estimates derived from the mean wind over the lowest 150 m of the wind sounding (LLM). Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC, and the solid vertical line corresponds to landfall.

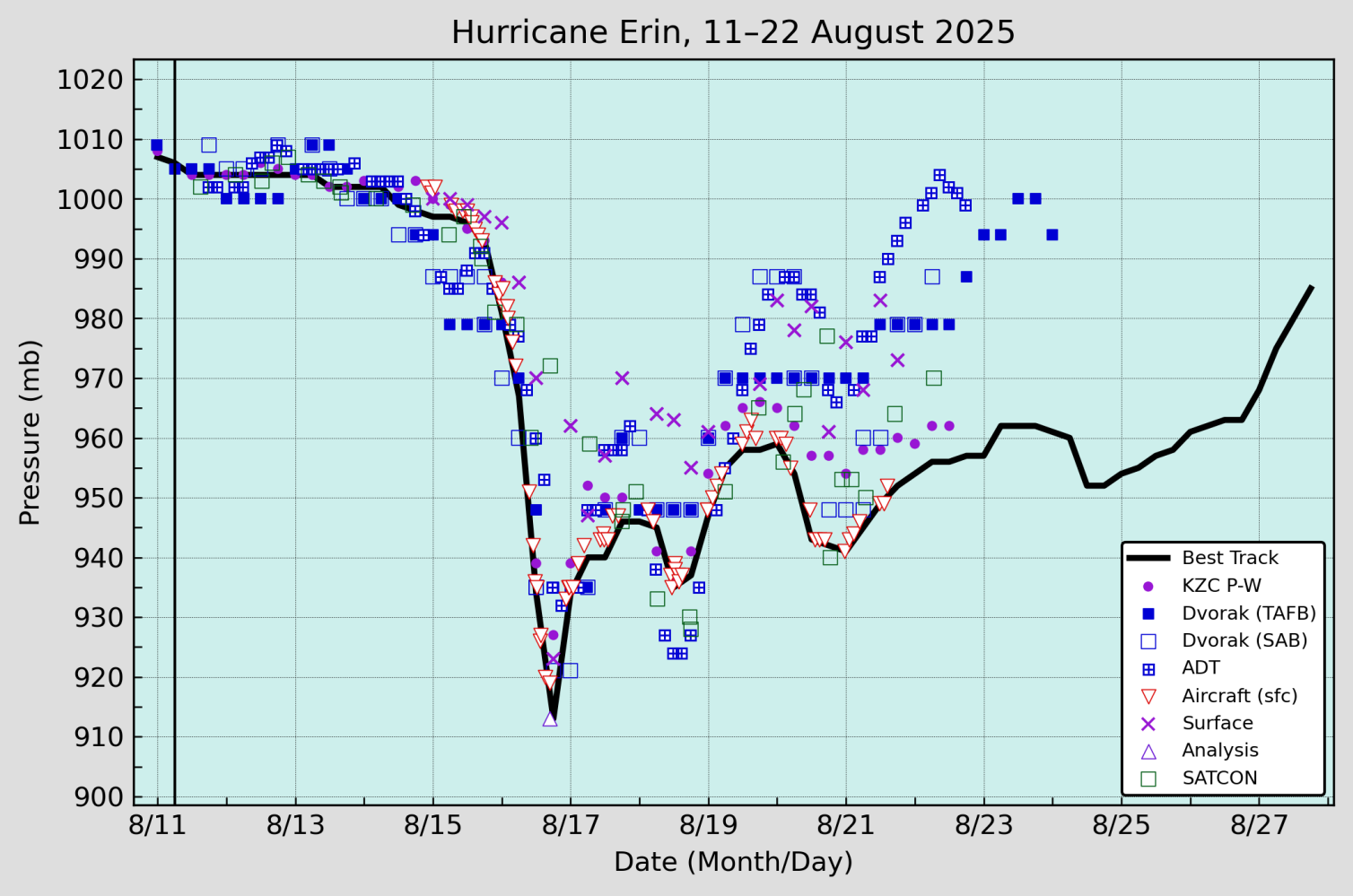


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Erin, 11–22 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC, and the solid vertical line corresponds to landfall.

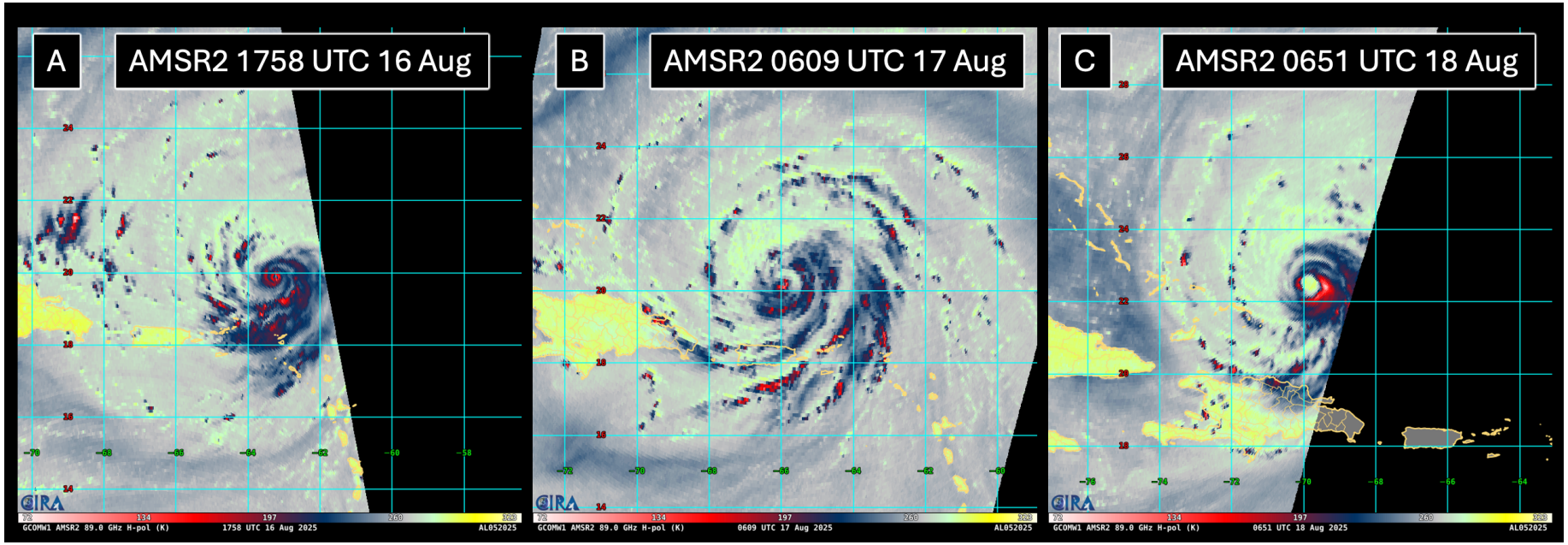


Figure 4. Microwave images showing Erin's structural evolution from 16–18 August. Panel (a) at 1758 UTC 16 Aug shows Erin near peak intensity with a very small eye. Panel (b) at 0609 UTC 17 Aug shows Erin undergoing an eyewall replacement with the inner eyewall decaying as outer rain bands develop around the smaller core. Panel (c) at 0651 UTC 18 Aug shows Erin completing an eyewall replacement with one single larger eyewall remaining.

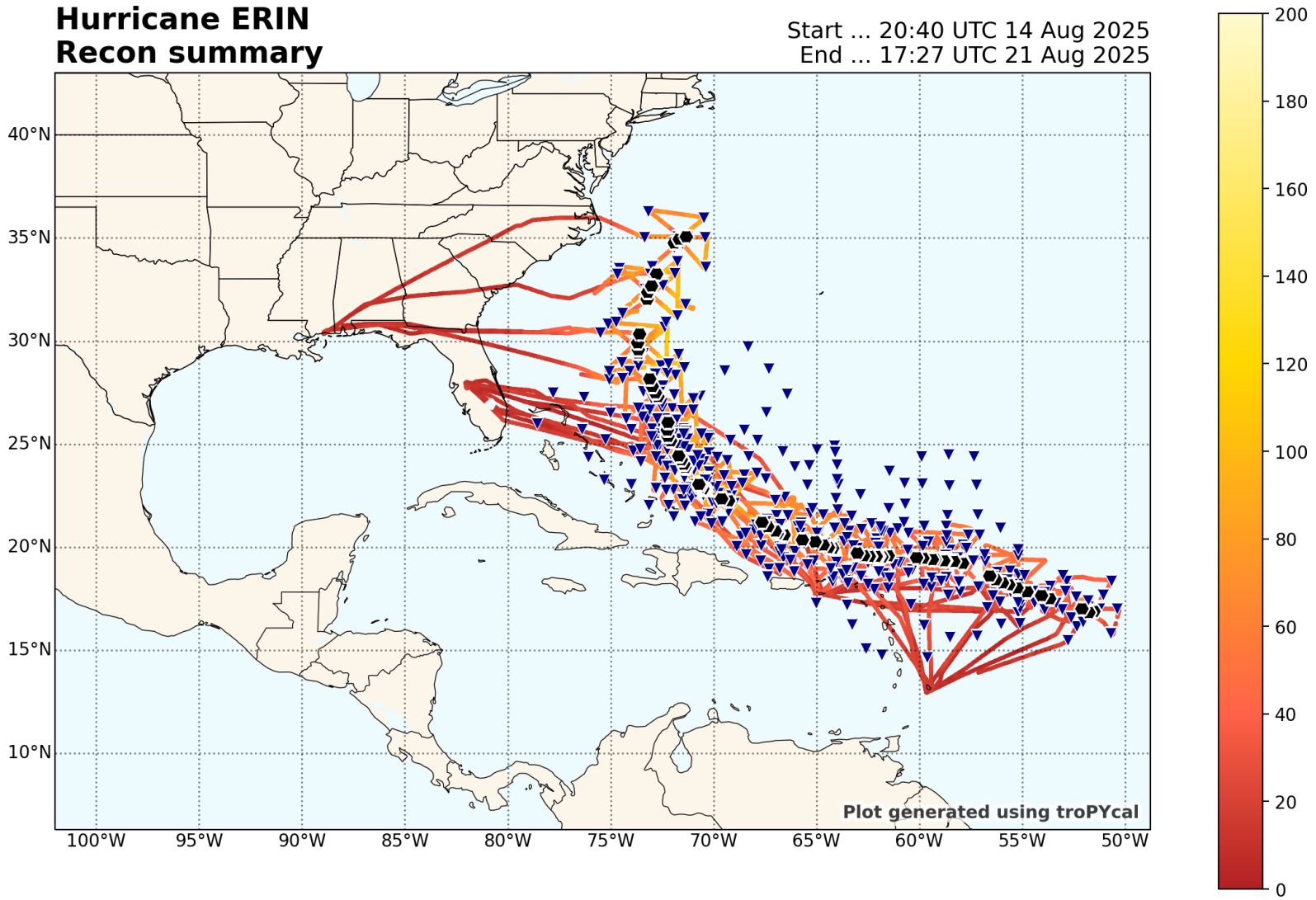


Figure 5. Hurricane Hunter aircraft flight tracks (colored lines) for missions into Hurricane Erin during 14–21, August, 2025. The black markers denote center fixes, and the blue triangles indicate dropsonde locations (including drops from synoptic surveillance missions, whose tracks are not shown). The color coding of the flight tracks represents the observed flight-level wind speed in knots at that location (see legend).

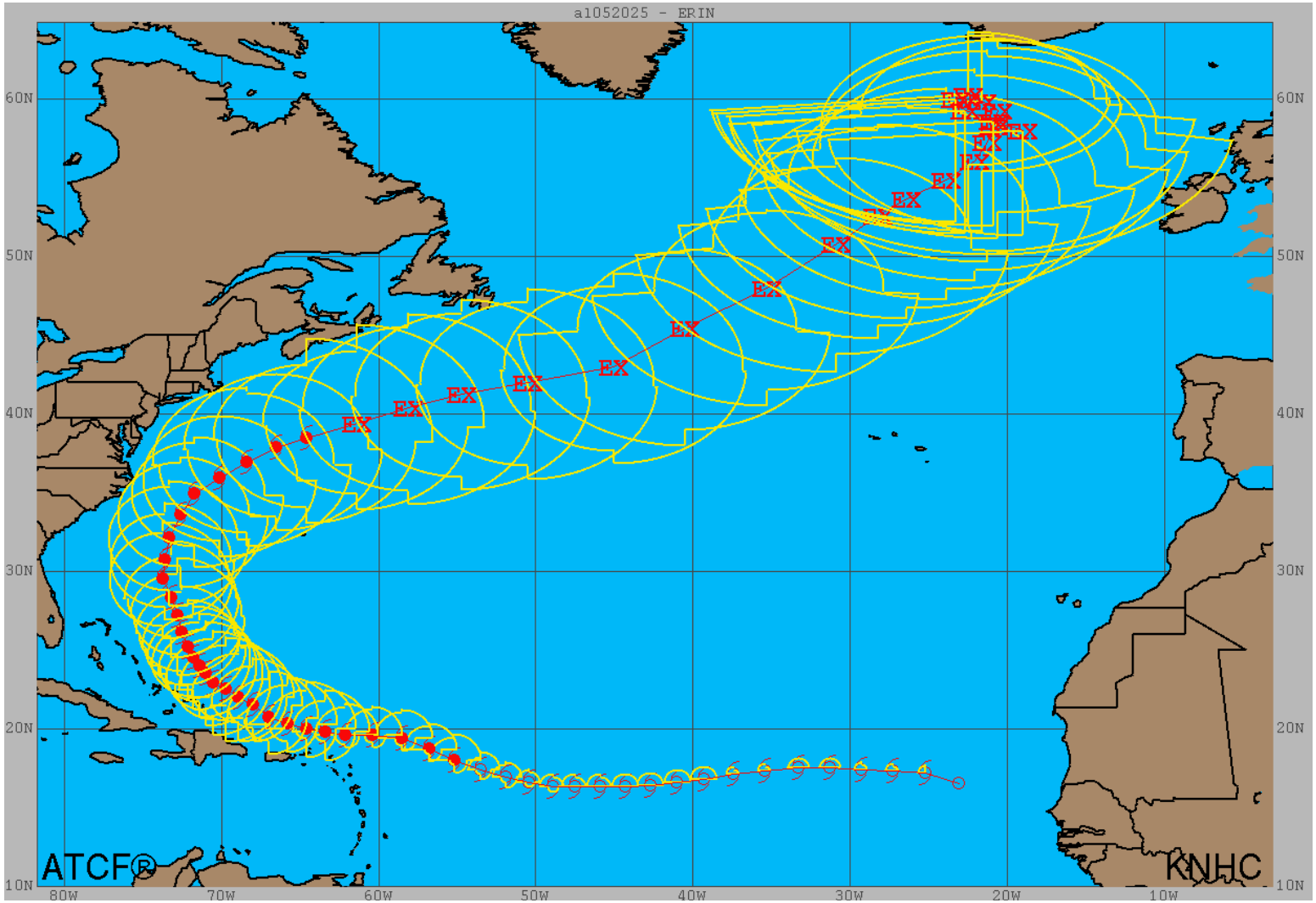


Figure 6. Extent of the 34-kt wind field during the lifetime of Hurricane Erin of 2025.

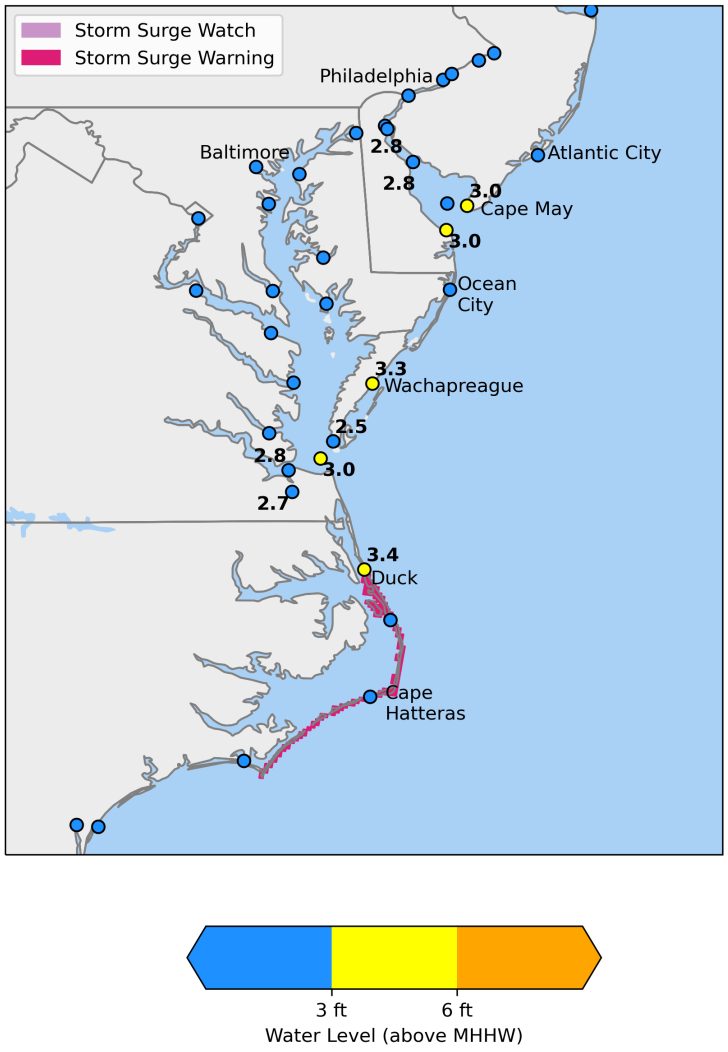


Figure 7. Storm surge observations and warnings for Hurricane Erin.

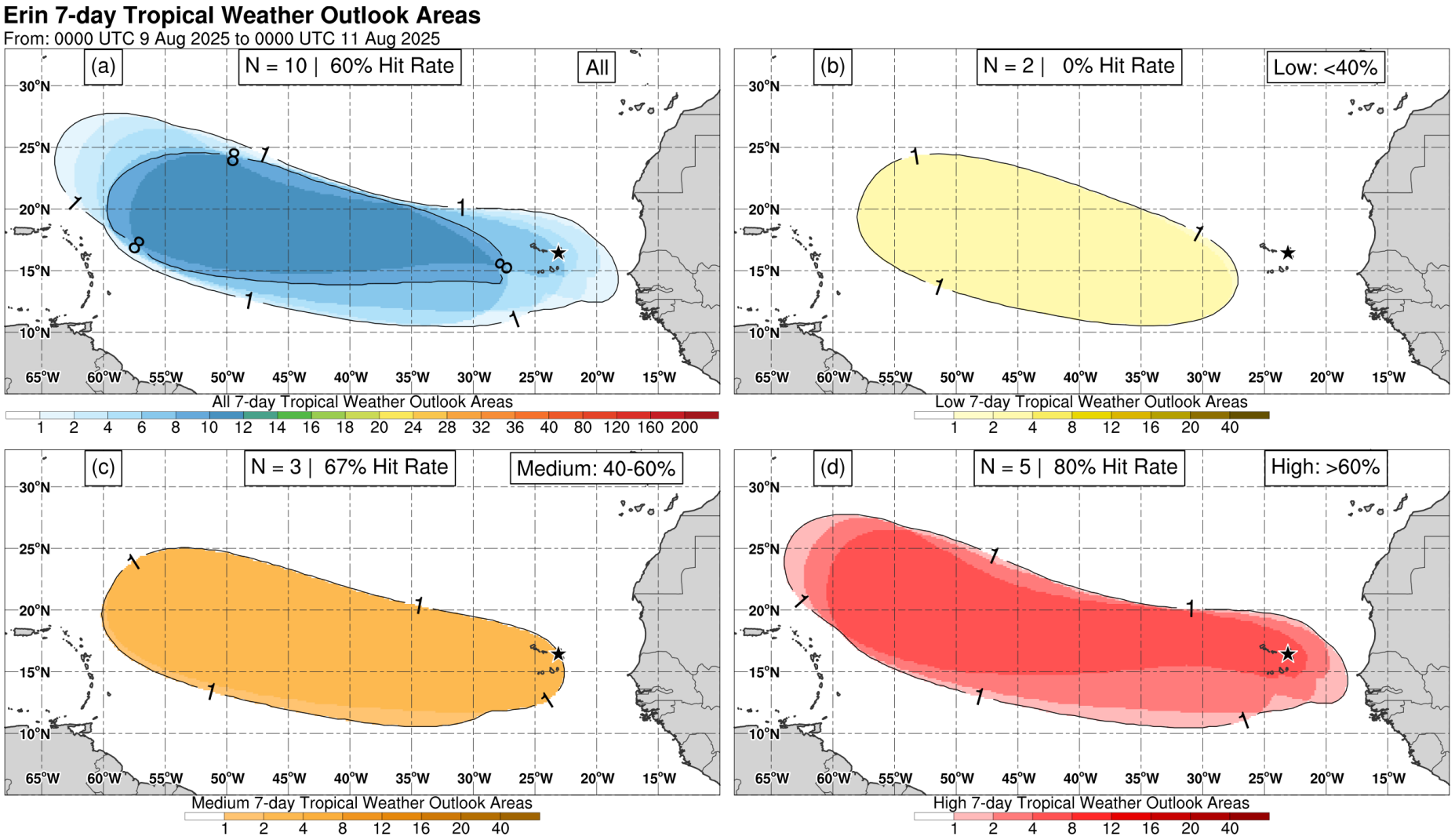


Figure 8. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Erin for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.

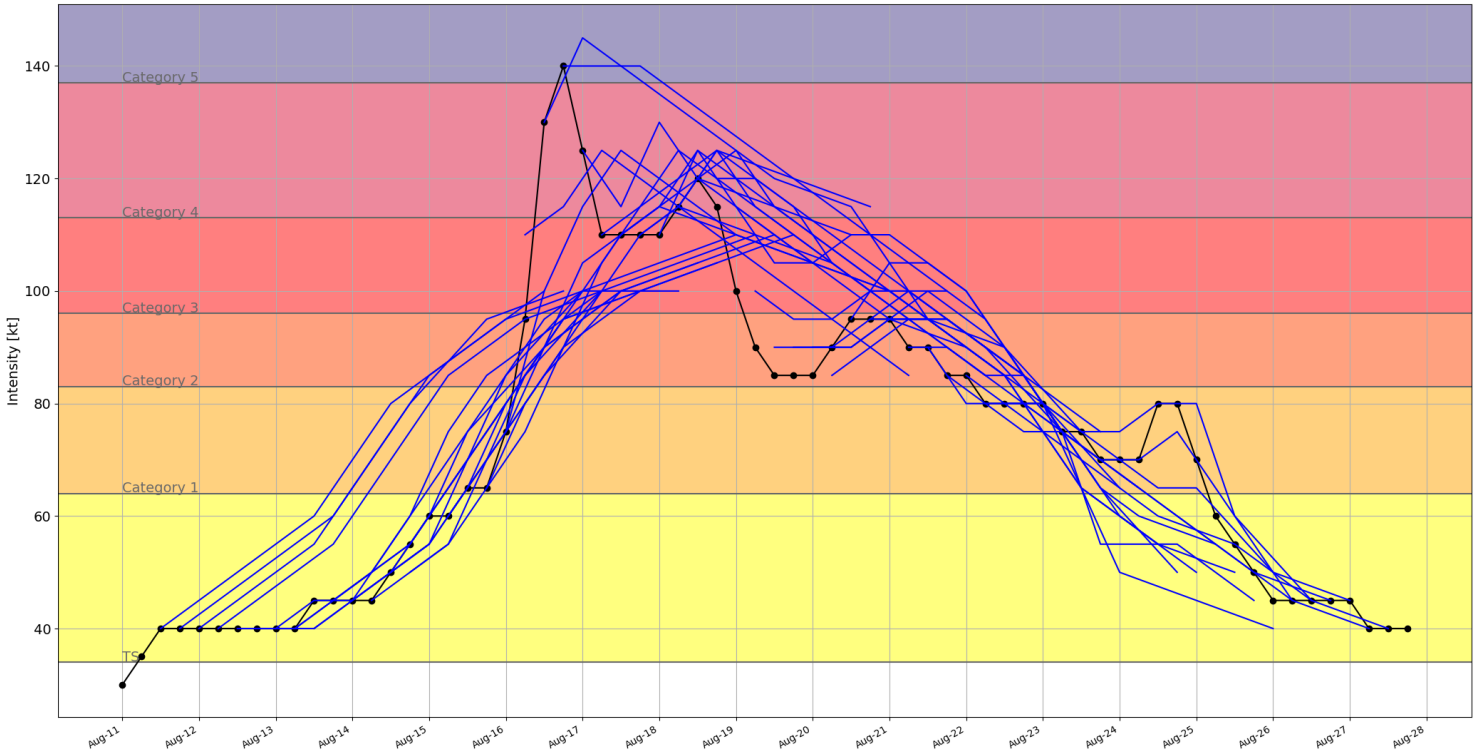


Figure 9. Official intensity forecasts for Erin from 11–22 August 2025.